

Unit 6

THE FIRM AND ITS EMPLOYEES

Video 1

If your boss isn't watching you, why not slack off?



Reservation Wage

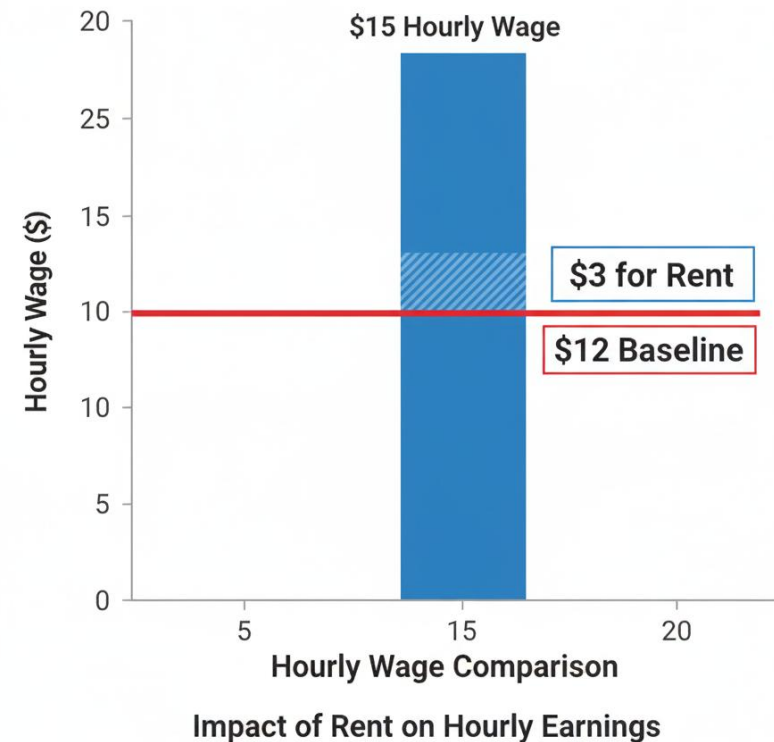
- *Definition:* lowest wage you'd accept for a job.
- *Example:* Would you take café job at \$12/hour?



Hourly Wage Comparison

Employment Rent

- *Definition:* extra benefit of having the job vs your next best alternative.
- *Example:* Café offers \$15/hour, reservation wage \$12 → rent = \$3/hour.



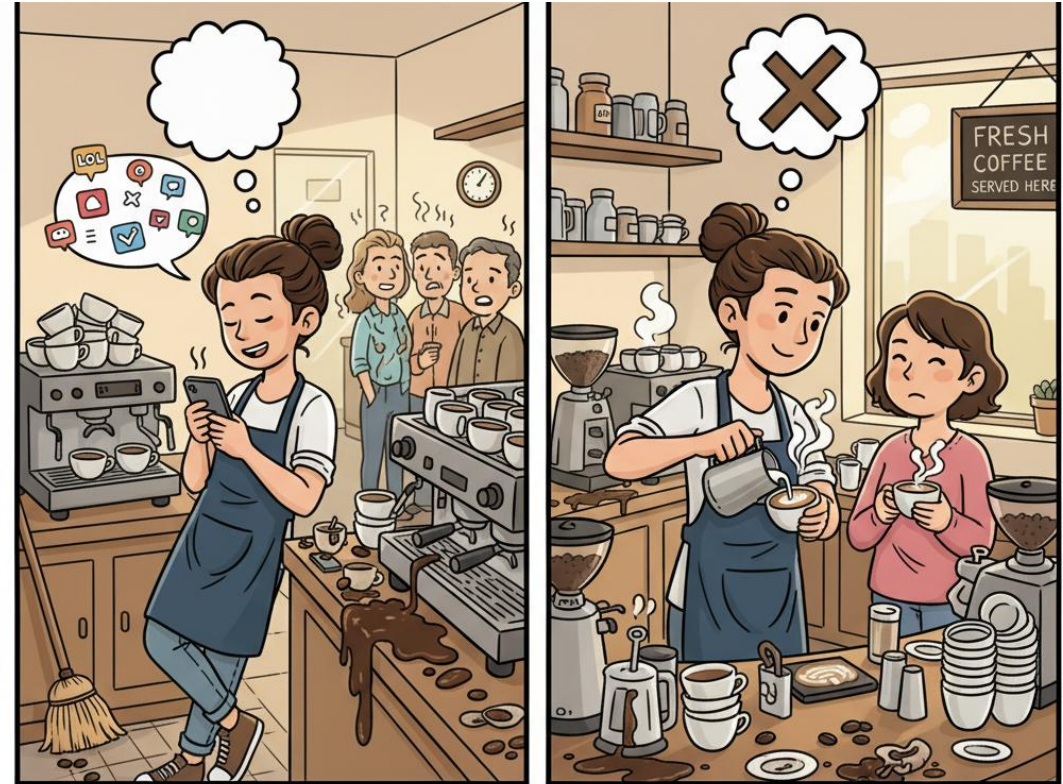
Principal–Agent Problem



- Boss (principal) wants effort; worker (agent) wants less effort.

Maria the Barista Scenario

- Work hard → keeps job.
- Slack off → risks losing job.
- Why work hard? → The value of the rent.



Wrap-up & Reflection

- Reservation wage = baseline.
- Employment rent = motivation.
- Firms set wages to balance effort vs cost.

Question: “Besides money, what makes people work hard?”

Video 2

Case Hook: Amazon Warehouses

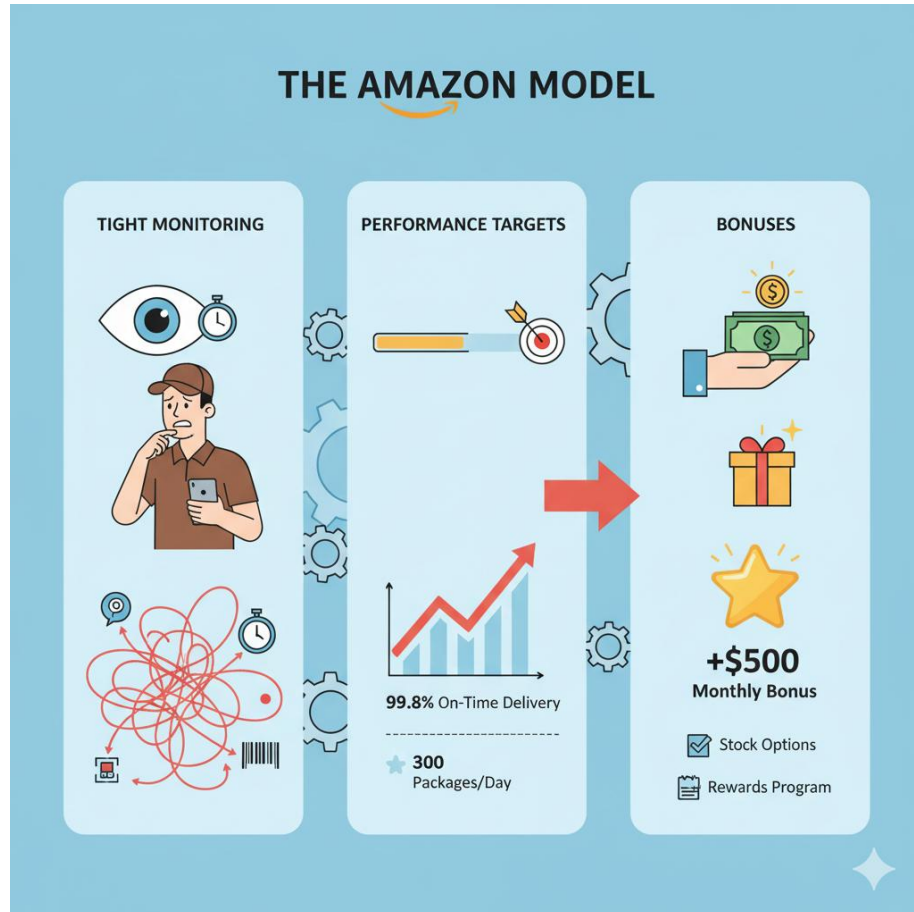
- Amazon employs over 1 million workers.
- Why do workers still work so hard, even under strict monitoring?



Theory Reminder

- Reservation wage = lowest acceptable wage
- Employment rent = value of job above alternatives
- Principal–agent = firm wants effort, worker wants comfort

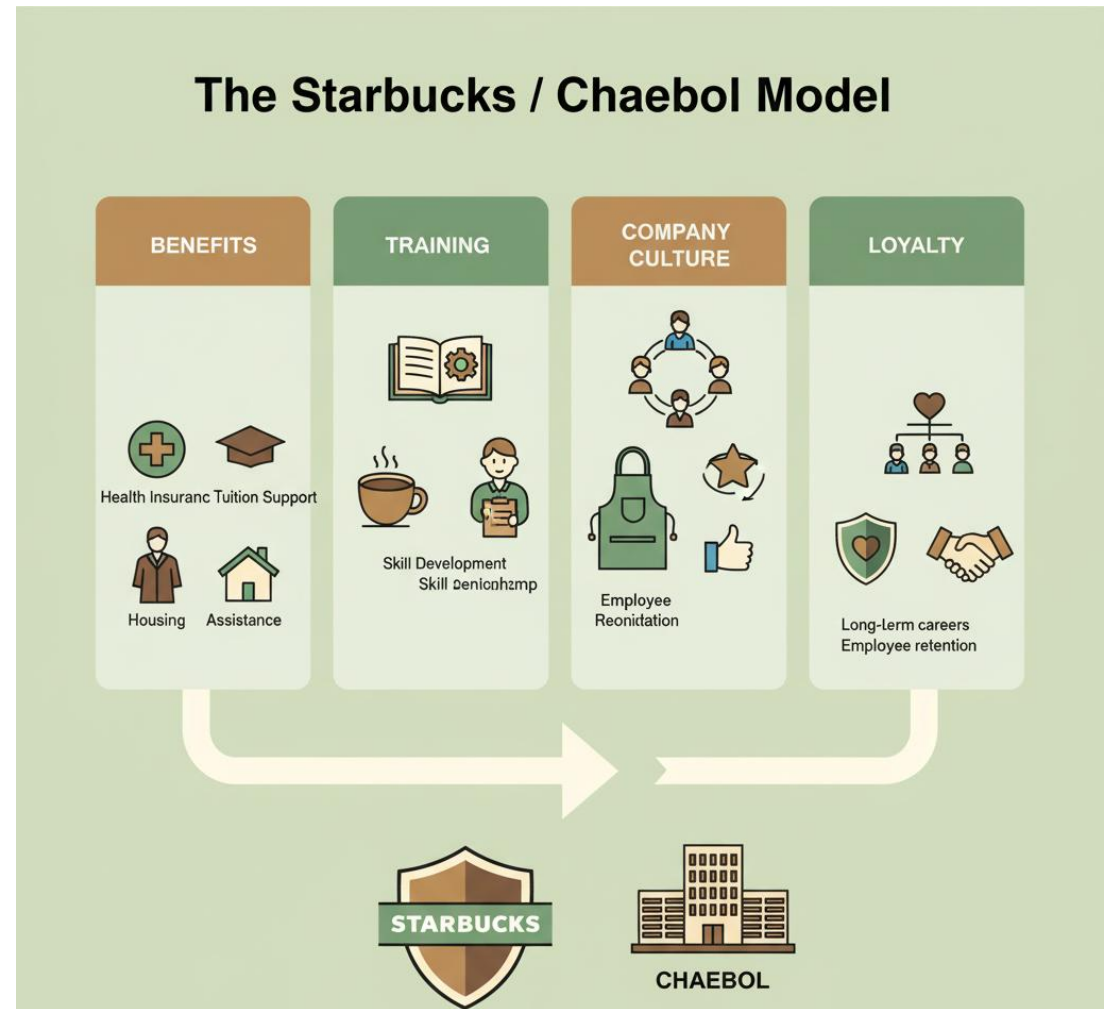
Firm Strategy 1: Amazon Model



- High wages & strict monitoring
- Productivity targets, surveillance
- Incentives through bonuses & penalties
- Some [critics](#)
- Ask AI with a *prompt* like “Amazon model: tight monitoring, performance targets, bonuses”

Firm Strategy 2: Starbucks / Chaebol Model

- Loyalty & culture
- Benefits, training, stability
- Motivation through pride & belonging
- Some [details](#)
- Ask AI with Prompt like
“Starbucks / chaebol model:
benefits, training, company
culture, loyalty”



Pause & Reflect

- Which strategy is more effective?
- *Stop the video and think of 2 reasons for your choice.*

Class Debate

- Position A: High wages + strict monitoring are essential.
- Position B: Non-monetary motivation (loyalty, pride) matters more.
- *Choose a side and prepare your argument.*

Wrap-Up & Assignment

- We saw theory applied to Amazon & Starbucks.
- Assignment: Write 150–200 words.
 - Option 1: Case analysis of employment rent & incentives
 - Option 2: Mini-debate, choose Position A or B
- Submit on Cyber Campus.

TODAY'S CLASS

1. Quiz review
2. Concepts and instructions
3. Diagnose the employment problem
4. Calculate employment rent
5. Design a management strategy
6. Peer review
7. Respond to an economic shock
8. Class debrief and submission

CONCEPTS FROM THE VIDEOS

- *Reservation wage*: The lowest wage at which a person is willing to accept a particular job.
- *Employment rent*: The net benefit of having the job compared with the worker's next-best alternative.
- *Incomplete contract*: A contract that cannot specify or enforce everything that matters to the employer and employee.
- *Principal–agent problem*: The employer wants the employee to act in the employer's interest, but cannot perfectly observe the employee's effort.

WHY IS THE EMPLOYMENT CONTRACT INCOMPLETE?

A contract may specify:

- Working hours
- Wage and benefits
- Job title
- Main responsibilities
- Measurable output targets

However, it cannot perfectly specify or observe:

- Effort
- Care and attention
- Cooperation
- Initiative
- Honesty
- Safe behaviour
- Quality of every decision

Today's problem: How can the firm motivate good performance when effort is difficult to observe?

THE CASE: GLOBALCART KOREA

- GlobalCart Korea processes international online orders.
- Employees receive goods, check product information, prepare packages, and arrange international shipment.

Managers can observe:

- The number of packages processed
- Attendance
- Whether employees meet formal targets

Managers cannot perfectly observe:

- Care and attention
- Cooperation with colleagues
- Product-handling quality
- Compliance with every safety procedure
- The employee's actual effort

MEET MARIA

Maria works in GlobalCart's international-order department.

- Weekly wage: KRW 700,000
- Value of benefits and job stability: KRW 80,000 per week
- Work-related costs: KRW 180,000 per week
- Net value of Maria's unemployment option: KRW 300,000 per week
- Expected time required to find a comparable job: 4 weeks
- Additional personal cost of high effort: KRW 100,000 per week
- Probability that shirking is detected: 5% per week

OUR SIMPLIFIED CLASSROOM MODEL

1. *Net benefit from the job:* $\text{Wage} + \text{non-wage benefits} - \text{work-related costs}$
2. *Weekly employment rent:* $\text{Net benefit from the job} - \text{net benefit from the reservation option}$
3. *Employment rent at risk:* $\text{Weekly employment rent} \times \text{expected weeks before finding a comparable job}$
4. *Expected penalty from shirking:* $\text{Probability of detection} \times \text{employment rent at risk}$

Prediction: Maria will choose high effort when the bonus for good performance plus the expected cost of being caught shirking is at least as large as the personal cost of working hard.

GROUP ORGANIZATION

Work in your assigned group.

Submit one report for the entire group.

Write the names and student numbers of all participating members.

Everyone must be able to explain the group's conclusions.

TASK 1: DIAGNOSE THE PROBLEM

Discuss the following questions:

1. Who is the principal?
2. Who is the agent?
3. What does the principal want?
4. What does the agent care about?
5. Which aspects of Maria's performance are difficult to observe?
6. Why can GlobalCart not solve the problem simply by writing a more detailed contract?

Write brief answers in your group report.

Time: 10 minutes

TASK 2: CALCULATE AND PREDICT

Calculate:

1. Maria's weekly net benefit from the job
2. Maria's weekly employment rent
3. Maria's total employment rent at risk
4. The expected penalty from shirking
5. Maria's reservation wage

Then answer:

6. Does the simplified model predict that Maria will choose high effort?
7. Explain the economic logic behind your prediction.

Show your calculations.

Time: 8 minutes

MANAGEMENT OPTIONS

OPTION A: HIGHER WAGE

GlobalCart increases Maria's weekly wage by KRW 100,000.

Cost to the firm: KRW 100,000 per week

Other conditions remain unchanged.

OPTION B: BETTER MONITORING

GlobalCart introduces improved monitoring.

Probability of detecting shirking:
Increases from 5% to 15%

Cost to the firm: KRW 40,000 per
employee per week

Other conditions remain unchanged.

MANAGEMENT OPTIONS

OPTION C: TRAINING AND BENEFITS

GlobalCart provides additional training and benefits.

Cost to the firm: KRW 60,000 per week

Value to Maria: KRW 120,000 per week

Maria's personal cost of high effort: Falls from KRW 100,000 to KRW 80,000 per week

OPTION D: PERFORMANCE BONUS

Maria receives a bonus of KRW 70,000 when she meets the performance target.

The bonus is paid only when the target is achieved.

All other conditions remain unchanged.

TASK 3: CHOOSE A MANAGEMENT STRATEGY

Evaluate all four options.

Your recommendation must explain:

For each option:

1. Calculate the new incentive to choose high effort.
2. Decide whether the option is sufficient in our simplified model.
3. Identify one possible disadvantage.

- Why the strategy should work
- Its cost to the firm
- One possible unintended consequence
- Why it is preferable to the alternatives

Then choose the best option for GlobalCart.

Time: 10 minutes

THINK BEYOND THE CALCULATION

Possible unintended consequences include:

- Employee stress
- Loss of trust
- Invasion of privacy
- Lower job satisfaction
- Higher employee turnover
- Excessive focus on measurable targets
- Neglect of quality or safety
- Manipulation of performance indicators
- Unequal effects on different employees

A strategy can produce the desired numerical incentive and still create other problems.

PEER REVIEW

Exchange your report with another group.

Review their recommendation.

Check:

1. Are the calculations correct?
2. Does the recommendation follow from the calculations?
3. Did the group consider the cost to the firm?
4. Is the identified disadvantage realistic?
5. Did the group overlook a better alternative?

Write:

- One strength of the recommendation
- One question or criticism
- One suggested improvement

Return the report to its original group.

Time: 7 minutes

NEW ECONOMIC CONDITIONS

SCENARIO A: STRONG LABOUR MARKET

Net value of unemployment option: Increases to KRW 450,000 per week

Expected time to find a comparable job: Falls to 1 week

SCENARIO B: RECESSION

Net value of unemployment option: Falls to KRW 200,000 per week

Expected time to find a comparable job: Increases to 8 weeks

NEW ECONOMIC CONDITIONS

SCENARIO C: BETTER MONITORING TECHNOLOGY

Probability of detecting shirking:
Increases from 5% to 20%

All other conditions remain unchanged.

SCENARIO D: IMPROVED WORKPLACE

Work-related costs: Fall from KRW 180,000 to KRW 120,000 per week

Personal cost of high effort: Falls from KRW 100,000 to KRW 70,000 per week

All other conditions remain unchanged.

TASK 4: RESPOND TO THE SHOCK

Your group will receive one scenario: A, B, C, or D.

Calculate how the scenario changes:

1. Maria's weekly employment rent
2. Maria's employment rent at risk
3. The expected penalty from shirking

Then discuss:

4. Is Maria now more or less likely to choose high effort?
5. Does the reservation wage change?
6. Does the wage required to discourage shirking increase or decrease?
7. Would you keep your original management strategy?

Prepare a one-minute explanation.

Time: 10 minutes

ONE-MINUTE GROUP REPORT

Our group analysed Scenario _____.

The scenario changes Maria's employment rent from _____ to _____.

The expected penalty from shirking changes from _____ to _____.

Maria is now more / less likely to choose high effort because:

The wage required to discourage shirking should increase / decrease / remain unchanged.

We would keep / change our original strategy because:

EMPLOYMENT RENT AND WORKER EFFORT

- The firm cannot perfectly observe or contract on effort.
- The possibility of losing employment rent can motivate employees to work harder.
- A stronger outside option reduces employment rent and weakens the cost of losing the job.
- A recession can increase employment rent because finding another comparable job becomes more difficult.
- Better monitoring makes shirking more likely to be detected.
- Better working conditions can increase employment rent and reduce the personal cost of effort.
- Wages, monitoring, benefits, and workplace culture are alternative—but imperfect—ways of addressing the principal–agent problem.

RESERVATION WAGE AND NO-SHIRKING WAGE

- Do not confuse these two ideas.
- Reservation wage
 - The minimum wage at which the employee is willing to accept the job.
- No-shirking wage
 - The wage required to make high effort preferable to shirking.
- Better monitoring may reduce the no-shirking wage without changing the reservation wage.
- A better outside option normally increases both the reservation wage and the wage required to discourage shirking.

SUBMISSION

Before submitting, confirm that your report includes:

- Names and student numbers
- Answers to Task 1
- All Task 2 calculations
- Evaluation of the four management options
- Final recommendation
- Peer-review comments
- Shock-scenario calculations
- Revised recommendation

Submit one report (pdf or Word file) per group.

Assessment:

10 points — Meaningful completion and participation

0 points — No submission or no meaningful completion

OUTLINE

A. Introduction

B. Structure of the firm

C. Hiring, quitting, and getting work done

D. Employment rent and reservation wage

E. The labour discipline and wage setting models

A. Introduction

The Context for This Unit

A firm's decision on the wage to offer workers and the number of workers to hire is another example of a constrained choice problem. (Unit 3)

The employer-employee relationship is an example of a strategic interaction. (Unit 4)

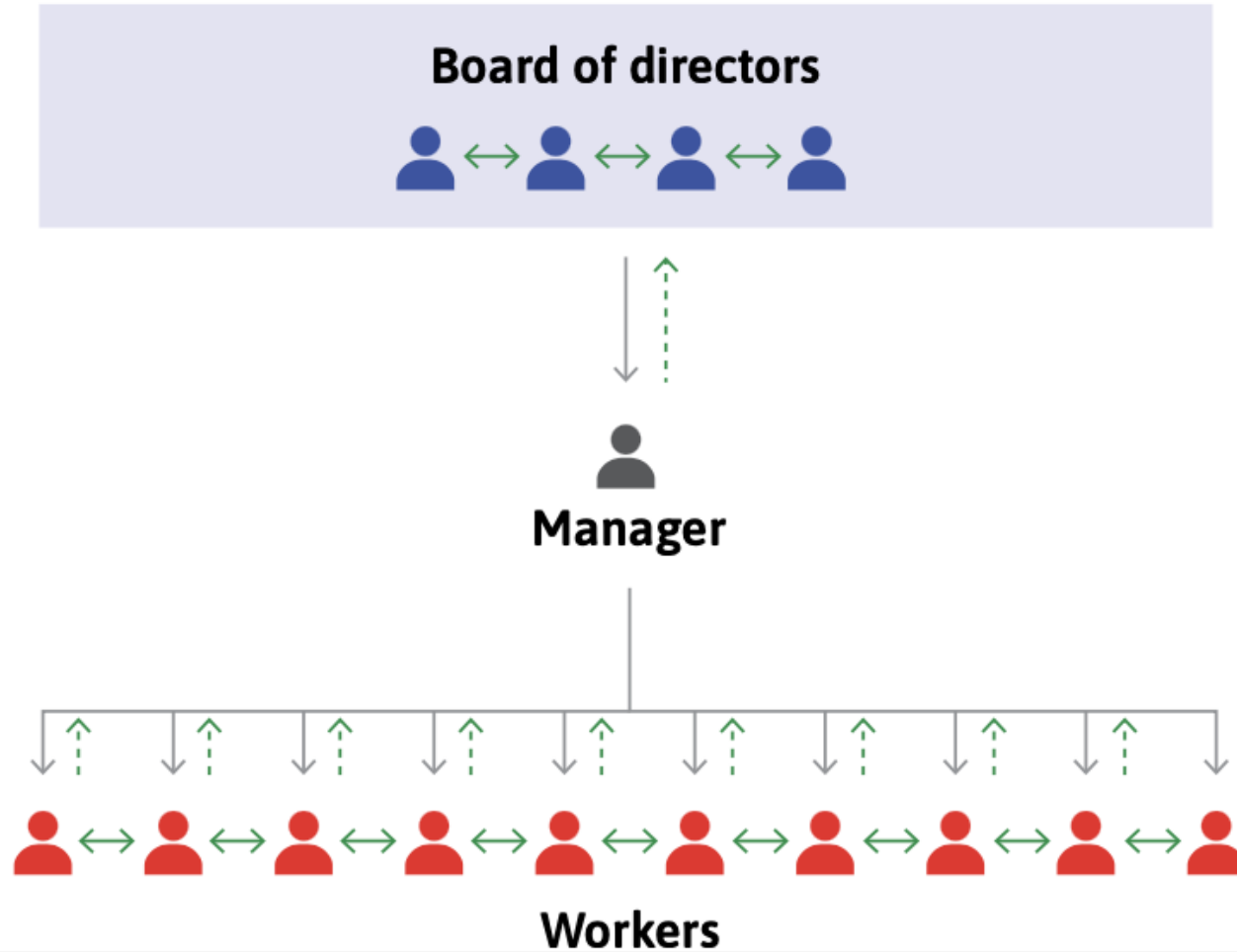
How do firms decide on the wage to set?
How does the firm-worker interaction affect wages and employment in the whole economy?

This Unit

- Describe key differences between contracts in firms and 'contracts' in markets.
- Explain the importance of reservation wages and employment rent.
- Discuss the principal-agent problem in this context.
- Show how the labour discipline and wage-setting models are used to find wages, worker effort, and employment levels.

B. Structure of the firm

Structure of the firm



Product vs labour contracts

Product

- Permanently transfer ownership of the good from seller to buyer.
- Our interactions with individual sellers or buyers are short-lived and often not repeated.

Labour

- Temporarily transfer *authority* over a person's activities from the employee to the manager or owner.
- An employment contract typically establishes a *long-term relationship* that may last years, decades, or even a lifetime.

Separation of ownership and control

- The firm's profits legally belong to the people who own the firm's **capital goods** and other **assets**.
- The owners take whatever remains after revenues are used to pay employees, managers, suppliers, creditors, and taxes. Profit is the *residual*. It is what's left of the revenues after these payments. The owners are called **residual claimants**. Managers and employees are not residual claimants.
- In small enterprises, the owners may also be the managers who take operational and strategic decisions. The owner-manager will usually try to maximize the profits of the enterprise.

Separation of ownership and control

- Large corporations usually have many owners. They are individuals and institutions that own the **shares** issued by the firm. Strategic and operational decisions are left to a relatively small group of specialized managers.
- When managers decide on the use of other people's funds, this is referred to as the **separation of ownership and control**.
- The owners receive the profits while the managers receive salaries, so it is not always in the interest of managers to maximize profits.

Separation of ownership and control

- There are two ways that owners can incentivize managers to serve their interests:
 - They can design contracts so that managerial compensation depends on the firm's share price.
 - Secondly, the board of directors, who represent the shareholders and typically have a substantial share in the firm themselves, can monitor managerial performance.

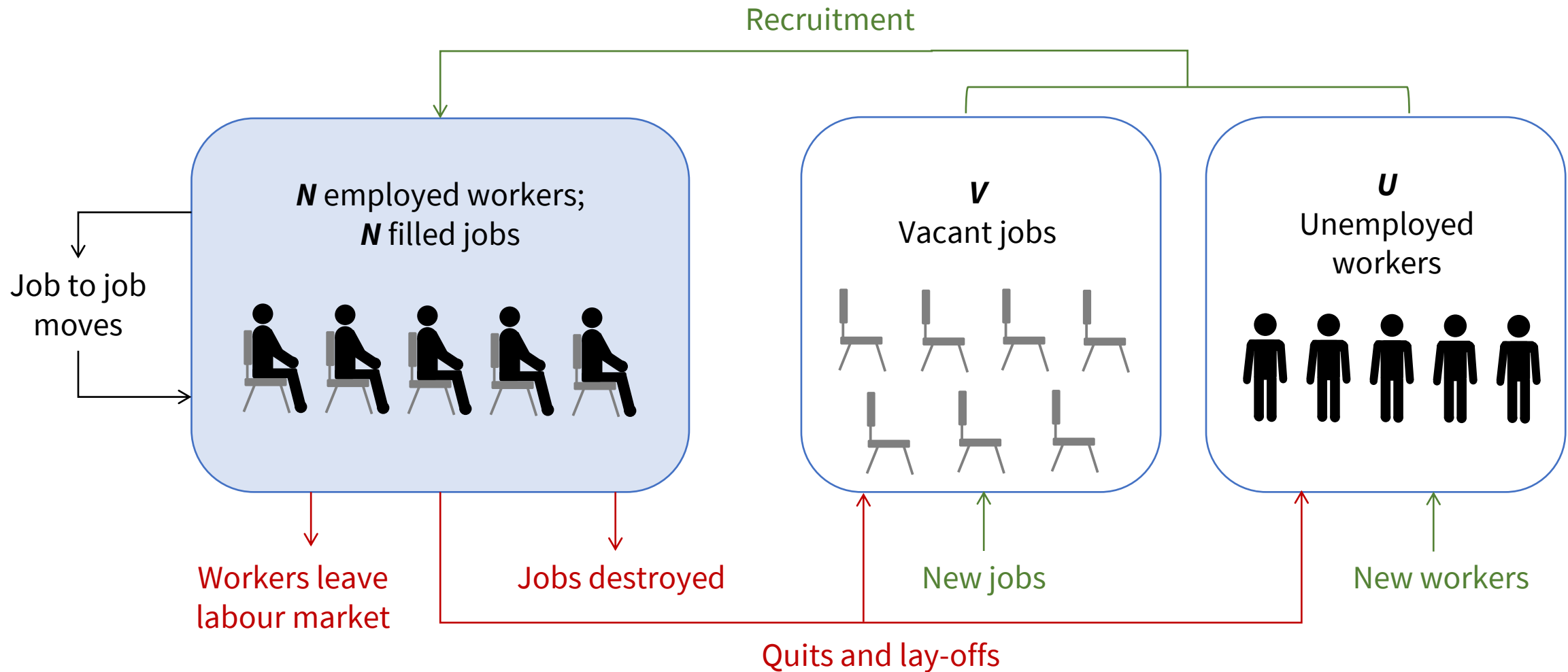
More on why labour markets are different

- The importance of finding a good match between worker and job.
- The costs for workers and employers of turnover.
- As the employment relationship progresses, the worker or manager can develop the skills needed to do the job well.
- The manager learns more about the worker's strengths and weaknesses.
- Social relationships develop too: some of our workmates will become our friends.

More on why labour markets are different

- These skills, networks, and friendships are referred to as **relationship-specific** or **firm-specific assets** because they are valuable only while an employee remains in a particular firm.
- The labour market is a **matching market**: people on each side of the market care about who they match with on the other side.

Labour market flows

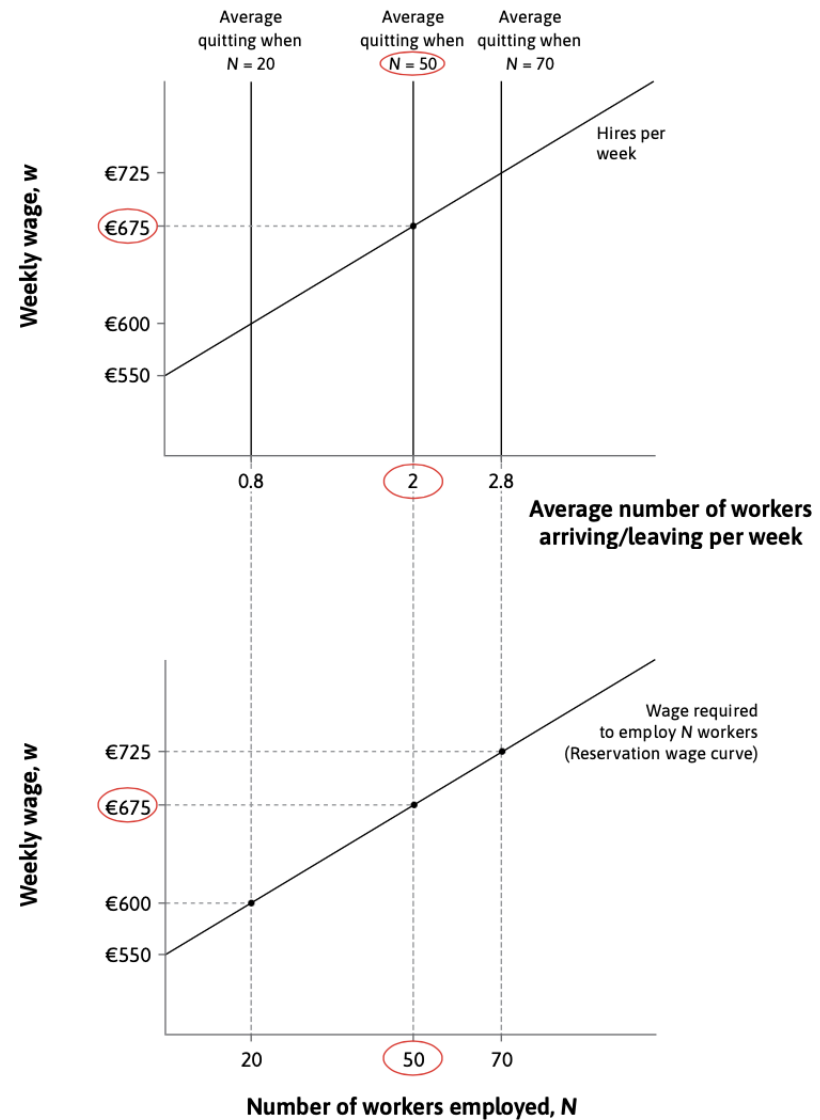


C. Hiring, quitting and getting work done

Reservation option and wage

- For an unemployed person their **reservation option**, or next-best alternative, is to remain unemployed and search for a job at a higher wage.
- The value of the reservation option depends on three things:
 - Her income while unemployed
 - Other things that affect her **utility** while she is unemployed
 - How long she thinks it would take to find a better opportunity.
- The minimum wage that an unemployed person would accept is called the **reservation wage**.

The reservation wage curve



Incomplete contracts

The problem of **incomplete contracts** arises because:

- The contract cannot specify exactly what the employee should do in every possible situation.
- It would be impractical or too costly for the firm to observe exactly how much effort each employee makes in every aspect of the job.
- Even if the firm could observe effort, it could not be the basis of a contract that would be enforceable in a court of law.
- The contract cannot prohibit the employee from quitting.

Principal-agent problem

- A game between two players, the **principal** and the **agent**, facing a conflict of interest.
- The employer is the principal and the worker is the agent.
- The principal would like to offer the agent an employment contract, and the agent wants the job.
- But there is a conflict of interest: to maximize the firm's profit the principal wants the agent to work hard; the agent prefers an easier life.
- In this employment example, the problem is a lack of information about exactly what the employee is doing.
- Information is either **asymmetric** or **non-verifiable** .

Solving the principal-agent problem

- Since effort cannot be specified in a contract, firms try to provide **incentives** for employees to work hard.
- One approach is to offer a contract in which pay depends, at least in part, on what the employee produces. This method of payment, in which pay depends entirely on what the individual produces, is known as **piece rate**.
- In modern knowledge- and service-based economies piece rates are impractical for most types of work.
- A widely-used alternative approach is to create incentives through **employment rents**.

D. Employment rent and reservation wage

Employment rent

- Workers care about keeping their jobs because the value of the job (taking into account all the benefits and costs it entails) is greater than the value of the next-best option.
- **Employment rent** = the difference between the net benefit from the job, and the net benefit of being unemployed.

Costs and benefits of working

There are some costs of working, such as:

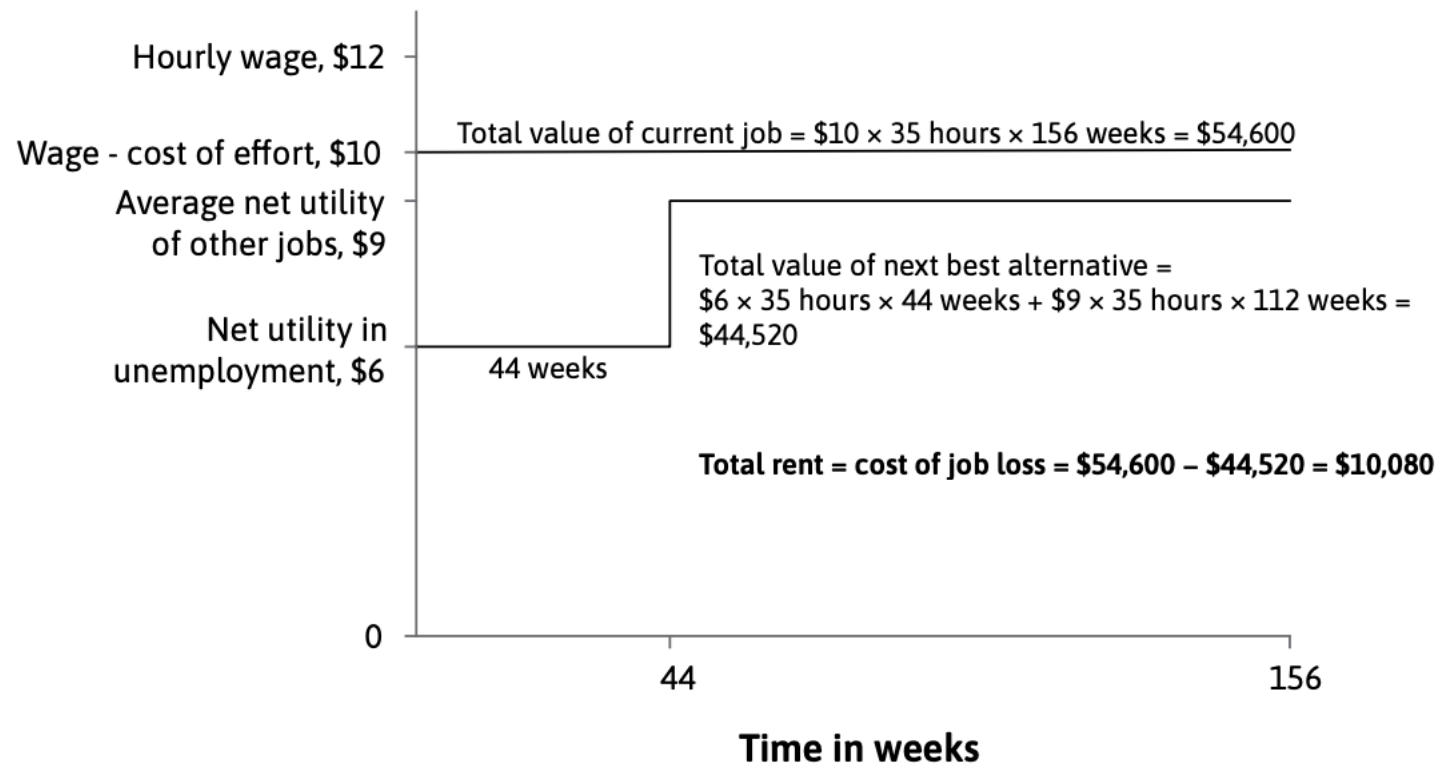
- The disutility of work
- The cost of travelling to work every day
- Childcare costs

There are many benefits of working, such as:

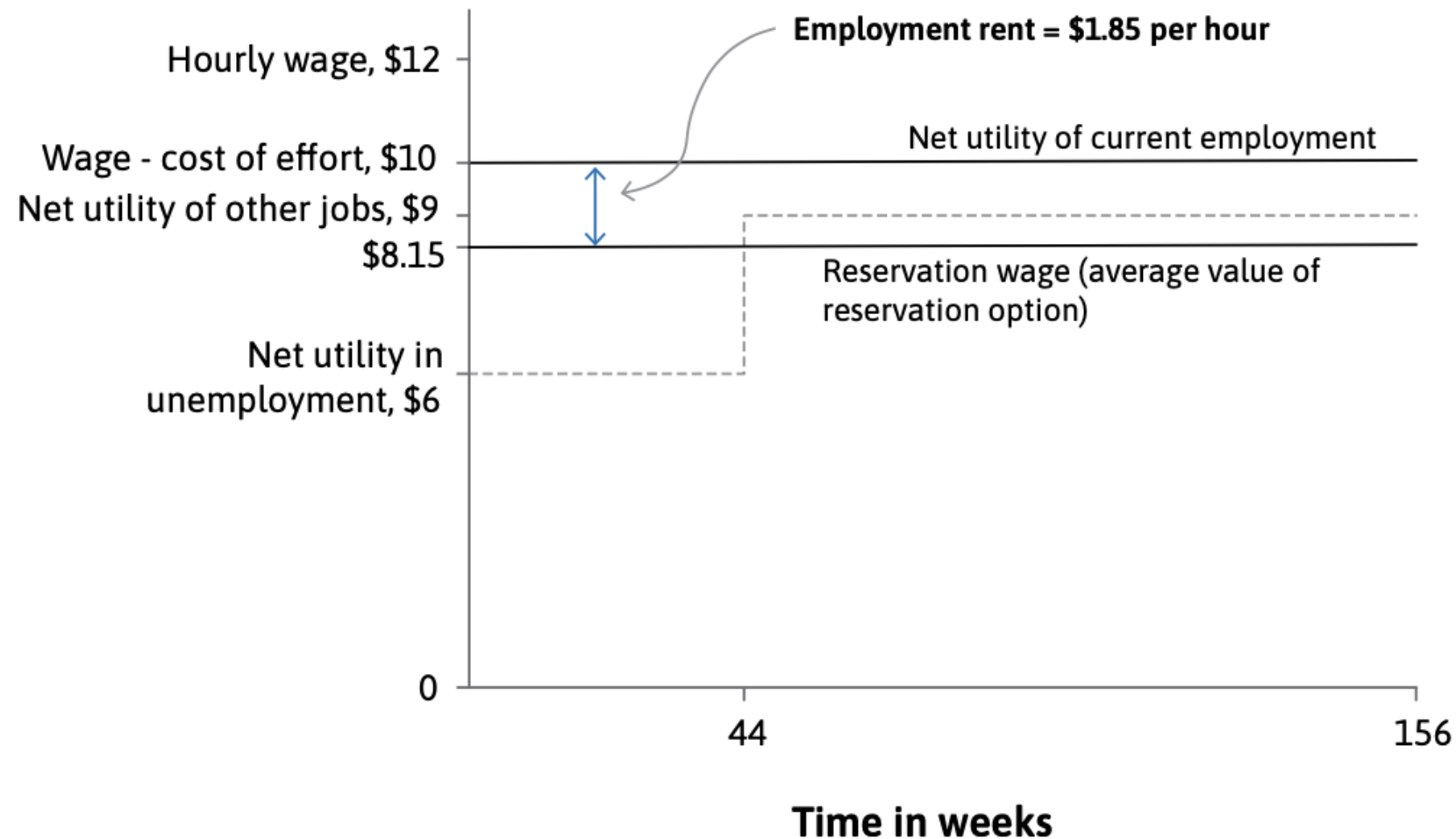
- Wage income
- Firm-specific assets
- Medical insurance
- Social status

Calculating employment rent

Calculating the employment rent received by Maria, an employee earning \$12 an hour for a 35-hour working week, when she exerts the amount of effort her employer requires:



Reservation wage



Reservation wage

- Her planning horizon is h weeks.
- If unemployed, her weekly unemployment benefit is b .
- Her additional net utility of being unemployed is α^M per week.
- The average net utility in other jobs (the wage minus the cost of effort) is v per week.
- She expects that it will take j weeks to find another job.
- $\tau (= j/h)$ is the proportion of time for which she can expect to remain unemployed.

$$w_r = \frac{j(b + \alpha^M) + (h - j)v}{h}$$

$$w_r = \tau(b + \alpha^M) + (1 - \tau)v$$

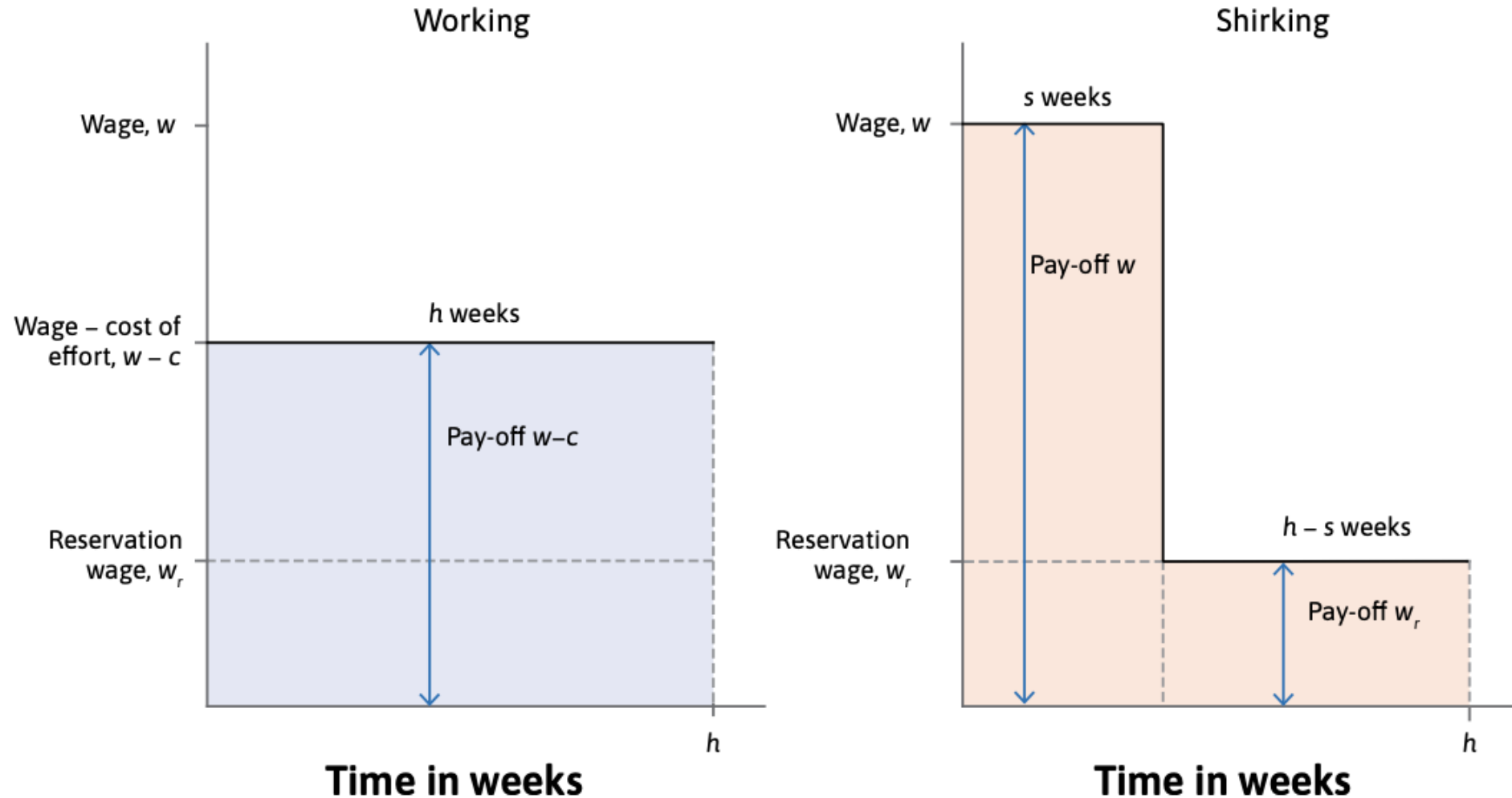
E. The labour discipline and wage-setting models

The labour discipline model

The players are Maria and her employer. The game is sequential (one of them chooses first). Here is the order of play:

1. Based on his knowledge of how employees like Maria respond to higher or lower wages, *the employer chooses a wage* and informs her that she will be employed in subsequent weeks at the same wage—as long as she works at the required level of effort.
2. In response to the wage offered and how likely it is that the employer will find out if she is not working up to the required level, *Maria chooses a level of work effort*, taking into account the costs of losing her job if she does not provide enough effort.

Maria's decision: work or shirk?



Maria's decision: work or shirk?

Maria's best response to the wage chosen by the employer is:

Work at required level of effort if:

Payoff from working hard for h weeks $\geq$ Payoff from shirking
(for s weeks until she is caught)

$$h(w - c) \geq sw + (h - s) w_r$$

Otherwise, make no effort.

The employer's decision: what wage?

Suppose that Maria produces output of y per week if she works at the required level, but zero if she does not. The firm's weekly profit is:

- $y - w$ if she works at the required level
- $-w$ if she shirks and has not been caught
- $-w$ if she is caught shirking and fired (she still has to be paid for her time on the job until she is fired)

So, the only way the employer can make a positive amount of profit is to set the wage high enough to ensure that Maria doesn't shirk, but below her output, y .

The employer's decision: what wage?

To maximize profit the employer should choose the lowest wage that gives her an incentive to work hard.

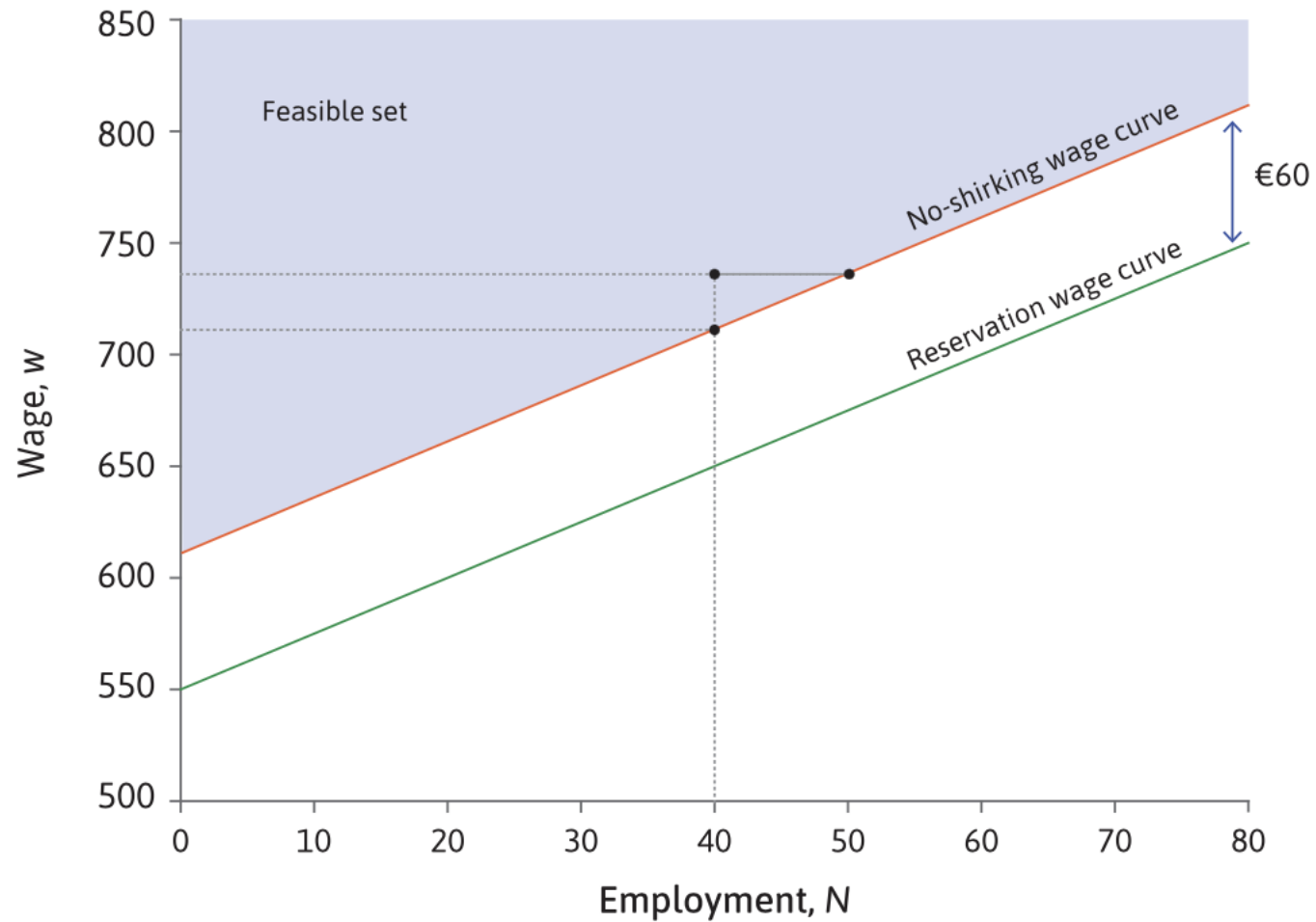
Maria's no-shirking wage:

$$w = w_r + c + \left(\frac{s}{h - s} \right) c$$

$$w = w_r + c + \text{rent}(s, c)$$

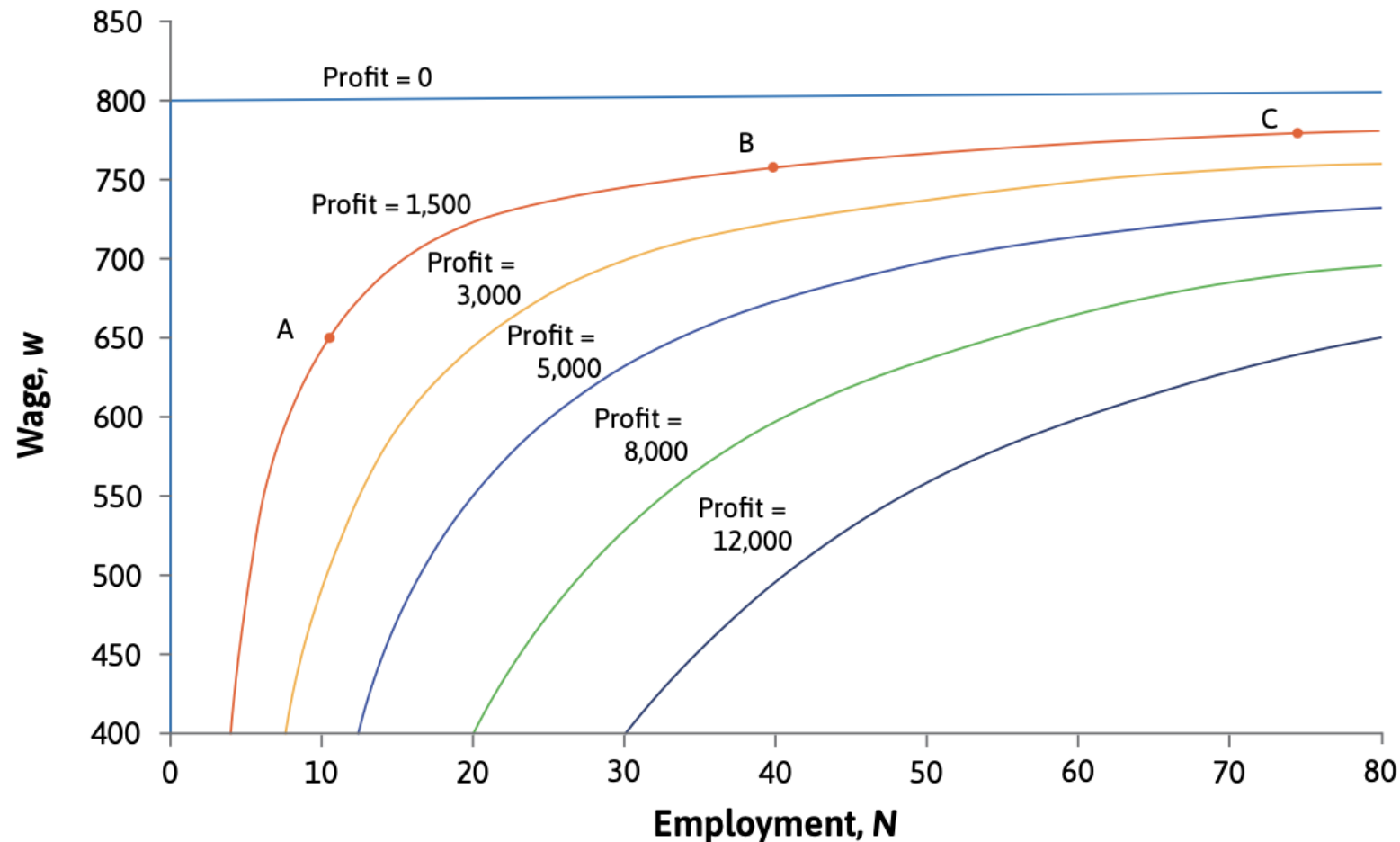
In the Nash equilibrium of the game, the employer sets the no-shirking wage and Maria chooses to exert effort.

The feasible set



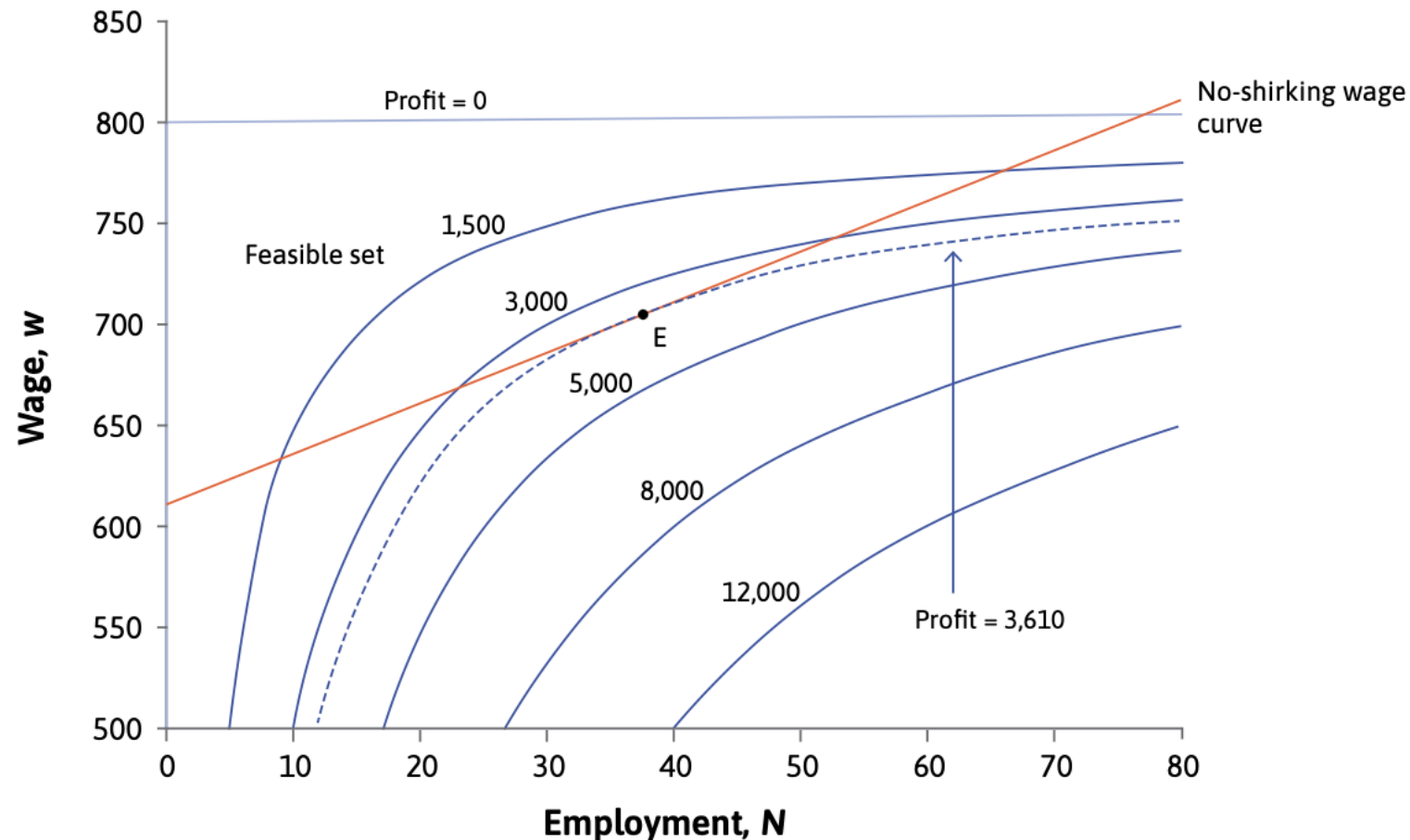
Isoprofit curves

Isoprofit curves connect all combinations of wages and employment that give the firm the same profit.



Profit maximisation

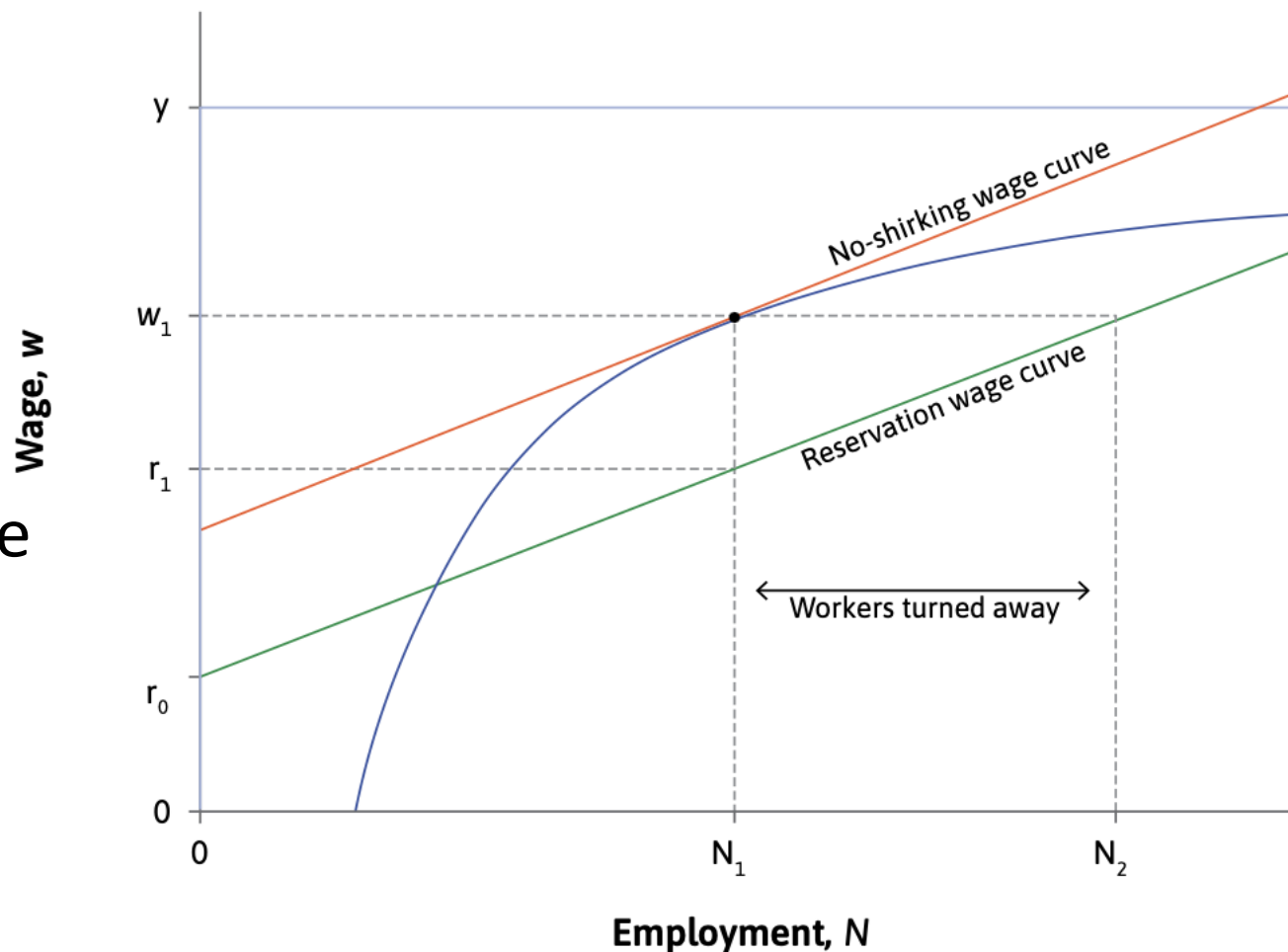
The firm's profit-maximising wage is where the no-shirking wage curve is tangent to an isoprofit curve.



Unemployment

Unemployment resulting from search and matching costs is known as frictional unemployment or **search unemployment**.

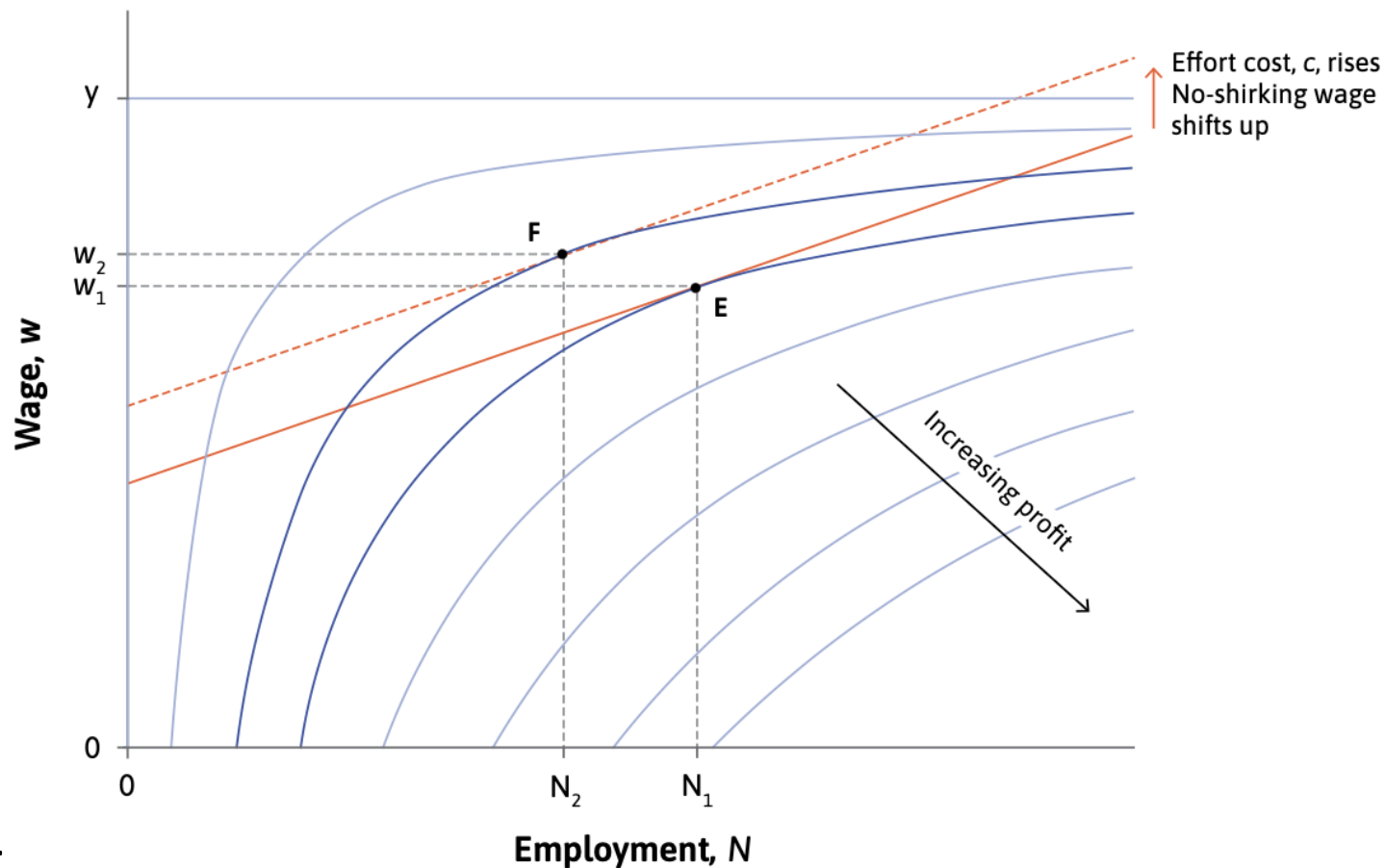
When firms set wages to provide incentives for effort, there must always be **involuntary unemployment**.



Shifts in the no-shirking wage curve

The equilibrium wage will change with:

- The firm's ability to detect shirking (s)
- Morale among employees (c)
- Competitors' demand for workers (v)
- Economic conditions (τ)
- The size of the unemployment benefit (b)



$$w = \tau \underbrace{(b + \alpha^N)}_{\text{reservation wage of } N^{\text{th}} \text{ worker}} + (1 - \tau)v + c + \text{rent}(s, c)$$

The whole economy

$$w = \tau(b + \alpha^N) + (1 - \tau)(w - c) + c + \text{rent}(s, c)$$

$$\Rightarrow \tau w = \tau(b + \alpha^N + c) + \text{rent}(s, c)$$

$$\Rightarrow w = b + \alpha^N + c + \frac{\text{rent}(s, c)}{\tau(u)}$$

For the whole economy, this equation is called the wage-setting or *WS* curve and represents the supply-side of the labour market.

Summary

- Distinctive features of employment: **wage-labour contract**; long-term relationships, **firm-specific assets**; **incomplete contracts**
- A model of hiring and quitting; **reservation wages** and the firm's **reservation wage curve**
- **Employment rent**; determining the reservation wage of an unemployed worker, the **labour discipline model** and the **no-shirking wage**
- The wage-setting model: **isoprofit curves**, and the **no-shirking wage curve**

In the next unit

- Model of interactions between customers and profit-maximising firms producing differentiated products
- Factors that affect the firm's choice of price and quantities produced (costs, **price elasticity**, **market power**)
- Surplus: measuring the gains from trade